University of Trier
Software Engineering Group
Prof. Dr. Stephan Diehl

*Information Visualization*
*Summer Semester 2026*
**Exercise Sheet 6**

*Group work:* Please keep in mind that the exercises are meant to prepare you for the exam.
*ChatBots:* Use your own brain as much and as often as possible.

## Exercise 1

Display the following data from the table below as a Scatterplot, as Parallel Coordinates and as a StarPlot. The diagrams must be drawn by hand and should not be automatically generated by R, Python or any other software.

| Name | Height in cm | Weight in kg | Speed in min/km |
|---|---|---|---|
| Alfred | 185 | 78 | 4,3 |
| Bernd | 180 | 90 | 6,3 |
| Cecilia | 160 | 55 | 5,0 |
| David | 190 | 85 | 5,2 |
| Elvira | 150 | 55 | 5,0 |

## Exercise 2

Python provides various packages for statistical analysis and to generate diagrams. Your task is to extend the Python script provided in StudIP so that it outputs the following descriptive statistics and visualizations for all columns of the dataframe stored in the variable `selection`.

1. Mean, median, quartiles (lower, upper), minimum, maximum
2. Boxplots for all columns in a single plot
3. Histograms for all columns in a single plot
4. Scatterplot matrix with all columns

## Useful Resources:

`https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.boxplot.html`
`https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.hist.html`
`https://pandas.pydata.org/docs/reference/api/pandas.plotting.scatter_matrix.html`